

PROLOG LABORATORY

Laboratory Project

on

AI Space Mission Simulation System

(Game Tree Generator Using Prolog)

Submitted

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ABSTRACT

The AI Space Mission Simulation System is an Artificial Intelligence based Game Tree Generator developed using Prolog. The project simulates decision-making processes during space mission crises by generating multiple future game states and outcomes.

The system uses recursive tree expansion to create branching mission paths based on user decisions. Different mission scenarios such as communication failure, asteroid collision, emergency rescue, and planetary exploration are represented as game tree structures.

The project demonstrates important Artificial Intelligence concepts such as state space representation, recursive search, decision tree generation, outcome evaluation, and intelligent recommendation systems.

The generated game tree helps users analyze possible mission outcomes, evaluate risks, and understand how AI systems simulate future decision paths.

INTRODUCTION

Artificial Intelligence plays an important role in modern gaming systems, decision support systems, and automated planning applications. One of the major concepts in AI is the Game Tree, which represents possible future states of a problem or game.

A Game Tree is a branching structure where each node represents a state and each branch represents a possible decision or move. AI systems use game trees to predict future outcomes and evaluate optimal decisions.

The AI Space Mission Simulation System is designed to demonstrate these concepts through an interactive sci-fi mission environment. The system generates mission decision trees and evaluates outcomes based on different user choices.

The project is implemented using Prolog, a logic programming language widely used in Artificial Intelligence applications.

PROBLEM DESCRIPTION

Space missions involve critical decision-making under uncertain conditions. A wrong decision may lead to mission failure, while an optimal decision can ensure survival and mission success.

The objective of this project is to design an AI-based Game Tree Generator that simulates multiple space mission scenarios and predicts possible outcomes using recursive tree expansion.

The system should:

- Generate mission decision trees
- Expand future mission states recursively
- Evaluate outcomes as safe or dangerous
- Recommend suitable mission actions
- Analyze mission risks
- Simulate intelligent decision-making

OBJECTIVES

The main objectives of the project are:

1. To implement a Game Tree Generator using Prolog.
2. To simulate AI-based mission decision-making.
3. To demonstrate recursive search and tree expansion.
4. To analyze mission outcomes and risks.
5. To provide AI recommendations for mission survival.
6. To create an interactive narrative-based AI system.

AI CONCEPTS USED:

1. Game Tree Representation

The project uses tree structures to represent mission decisions and future states.

2. State Space Representation

Each mission situation is represented as a separate state in the AI system.

3. Recursive Search

The system recursively expands mission branches to generate future outcomes.

4. Decision Tree Generation

Possible mission choices are represented as decision branches.

5. Rule-Based Reasoning

The system uses Prolog rules to evaluate outcomes and recommendations.

6. Intelligent Outcome Evaluation

Mission endings are classified as safe, dangerous, or risky.

SYSTEM MODULES

Module 1 – Mission Initialization

Initializes the space mission scenario.

Module 2 – Decision Generator

Generates possible mission choices.

Module 3 – Recursive Tree Expansion

Expands future mission paths recursively.

Module 4 – Recommendation System

Suggests safe mission actions.

Module 5 – Risk Analyzer

Classifies mission risks as low, medium, or high.

Module 6 – Mission Status Evaluator

Analyzes final mission outcomes.

Module 7 – Tree Display System

Displays the complete AI game tree.

ALGORITHM

Step 1:

Start the AI mission simulation system.

Step 2:

Initialize the root mission scenario.

Step 3:

Generate possible mission decisions.

Step 4:

Expand each decision recursively into future outcomes.

Step 5:

Display the generated game tree structure.

Step 6:

Evaluate outcomes as safe or dangerous.

Step 7:

Analyze mission risk levels.

Step 8:

Provide AI-based mission recommendations.

Step 9:

Display mission status.

Step 10:

Stop.

PROLOG PROGRAM CODE

1. Root Scenario + Choice Generation:

```
start('Spaceship NOVA-X loses communication near Planet Zorath.').
```

```
choice(start, 'Repair Communication System').
```

```
choice(start, 'Explore Nearby Planet').
```

```
choice(start, 'Send Emergency Signal').
```

2. Recursive Tree Expansion Logic:

```
show_tree(start) :-
```

```
    display_tree(start, 0).
```

```
display_tree(Node, Depth) :-
```

```
    indent(Depth),
```

```
    write('|-- '),
```

```
    write(Node), nl,
```

```
    choice(Node, Next),
```

```
    NewDepth is Depth + 1,
```

```
    display_tree(Next, NewDepth),
```

```
    fail.
```

3. Outcome Evaluation Logic:

```
safe('Mission Successfully Saved').
```

```
danger('Spaceship Destroyed').
```

```
evaluate(Outcome) :-
```

```
    safe(Outcome),
```

```
    write('SAFE OUTCOME').
```

```
evaluate(Outcome) :-
```

```
    danger(Outcome),
```

```
    write('DANGEROUS OUTCOME').
```

4. AI Recommendation System:

```
recommend('Repair Communication System') :-  
    write('AI Recommendation: Repair Communication System is the safest option.').
```

5.Mission Status Analyzer Module:

```
mission_status('Mission Successfully Saved') :-  
    write('MISSION STATUS: SUCCESS'), nl.
```

```
mission_status('Crew Rescued Safely') :-  
    write('MISSION STATUS: SUCCESS'), nl.
```

```
mission_status('Spaceship Destroyed') :-  
    write('MISSION STATUS: FAILURE'), nl.
```

```
mission_status('Alien Fleet Attacks') :-  
    write('MISSION STATUS: FAILURE'), nl.
```

SAMPLE QUERIES AND OUTPUTS

Query 1: Full Game Tree

```
File Settings Tools Debug Help

40 ?- show_tree(start).

=====
AI SPACE MISSION SIMULATION SYSTEM
=====

Mission AI Initialized Successfully

|-- start
  |-- Repair Communication System
    |-- Restore Contact With Earth
      |-- Mission Successfully Saved
    |-- Trigger Reactor Failure
      |-- Spaceship Destroyed
  |-- Explore Nearby Planet
    |-- Find Alien Technology
      |-- Advanced Energy Source Acquired
    |-- Encounter Hostile Creature
      |-- Crew Members Injured
  |-- Send Emergency Signal
    |-- Rescue Ship Arrives
      |-- Crew Rescued Safely
    |-- Enemy Detects Signal
      |-- Alien Fleet Attacks
  |-- Handle Asteroid Collision
    |-- Activate Energy Shields
      |-- Asteroid Deflected Successfully
      |-- Shield System Failure
    |-- Change Space Route
      |-- Safe Alternate Route Found
      |-- Entered Dangerous Black Hole
  |-- Launch Escape Pods
    |-- Crew Escaped Safely
    |-- Pods Lost In Space

true.

40 ?- ▲
```

Query 2: Exploration Mission

```
File Settings Tools Debug Help

|-- Handle Asteroid Collision
  |-- Activate Energy Shields
    |-- Asteroid Deflected Successfully
    |-- Shield System Failure
  |-- Change Space Route
    |-- Safe Alternate Route Found
    |-- Entered Dangerous Black Hole
  |-- Launch Escape Pods
    |-- Crew Escaped Safely
    |-- Pods Lost In Space

true.

54 ?- start_mission(exploration).

===== PLANET EXPLORATION MISSION =====

|-- Explore Nearby Planet
  |-- Find Alien Technology
    |-- Advanced Energy Source Acquired
  |-- Encounter Hostile Creature
    |-- Crew Members Injured

true.

55 ?- ▲
```

Query 3: Asteroid Mission

```
File Settings Tools Debug Help

===== ASTEROID COLLISION MISSION =====

|-- Handle Asteroid Collision
|-- Activate Energy Shields
|-- Asteroid Deflected Successfully
|-- Shield System Failure
|-- Change Space Route
|-- Safe Alternate Route Found
|-- Entered Dangerous Black Hole
|-- Launch Escape Pods
|-- Crew Escaped Safely
|-- Pods Lost In Space

true.

56 ?- ▲
```

Query 4:Next Possible Choices

```
File Settings Tools Debug Help

|-- Safe Alternate Route Found
|-- Entered Dangerous Black Hole
|-- Launch Escape Pods
|-- Crew Escaped Safely
|-- Pods Lost In Space

true.

56 ?- next_choice('Explore Nearby Planet').
Possible Outcome: Find Alien Technology
Possible Outcome: Encounter Hostile Creature
true.

57 ?- ▲
```

Query 5:Outcome Evaluation & AI Recommendation

```
File Settings Tools Debug Help

true.

56 ?- next_choice('Explore Nearby Planet').
Possible Outcome: Find Alien Technology
Possible Outcome: Encounter Hostile Creature
true.

57 ?- evaluate('Alien Fleet Attacks').
Alien Fleet Attacks --> DANGEROUS OUTCOME
true.

58 ?- recommend('Handle Asteroid Collision').
AI Recommendation: Activate Energy Shields for maximum protection.
true.

59 ?- mission_status('Crew Rescued Safely').
MISSION STATUS: SUCCESS
true.

60 ?- risk_level('Explore Nearby Planet')
```

Query 6: Mission Status & Risk Analysis

```
File Settings Tools Debug Help
Possible Outcome: Encounter Hostile Creature
true.

57 ?- evaluate('Alien Fleet Attacks').
Alien Fleet Attacks --> DANGEROUS OUTCOME
true.

58 ?- recommend('Handle Asteroid Collision').
AI Recommendation: Activate Energy Shields for maximum protection.
true.

59 ?- mission_status('Crew Rescued Safely').
MISSION STATUS: SUCCESS
true.

60 ?- risk_level('Explore Nearby Planet').
RISK LEVEL: HIGH RISK
true.

61 ?-
```

ADVANTAGES:

- Demonstrates real AI concepts.
 - Uses recursive search and decision trees.
 - Interactive and visually attractive.
 - Supports multiple mission simulations.
 - Easy to expand with additional missions.
 - Provides intelligent recommendations.
-

APPLICATIONS:

- AI-based gaming systems
- Decision support systems
- Mission planning systems
- Simulation environments
- Educational AI demonstrations
- Strategic planning applications

REFERENCES

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